medsum-fact — Review 2 Baseline Report

Plain 4-bit QLoRA fine-tune of Llama-3.1-8B-Instruct for dialogue → clinical note

Run **baseline_qlora_llama31_8b** · mode **full** · Kaggle kernels `hirthick17/medsum-baseline-qlora` (train) → `hirthick17/medsum-baseline-infer` (infer) · generated 2026-09-19T11:43:24 from repo commit a7504b8

NOT THE BASELINE. trainer_state.json points at a *_smoke checkpoint but the config says mode=full — the training half of this run is the smoke run's.

1. Run summary

The baseline defined in REVIEW2_BASELINE.md: no fact table, no ORPO, no verifier, no provenance. One instruction-tuned decoder, 4-bit NF4 QLoRA SFT on the assistant tokens only, greedy inference on the six official test splits.

Field	Value
Base model	unsloth/Meta-Llama-3.1-8B-Instruct ungated mirror of meta-llama/Llama-3.1-8B-Instruct (Llama 3.1 Community Licence)
Quantization	4-bit NF4, double-quant=True, compute torch.float16
LoRA	r=16, α=32, dropout=0.05, targets: q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj trainable 41.9 M of 8.07 B (0.52 %)
Data	train 4265 · val 220 · max_seq_len 4096 mix: {"dialogue_soap": 3000, "mts_dialog": 1198, "aci_bench": 67} · leakage check: passed
Optimisation	lr 0.0002 · cosine · warmup 0.03 · paged_adamw_8bit · batch 1 × accum 16 · 5 epochs · seed 42 · liger fused-CE=True
Hardware (train)	2x Tesla T4 · 29.1 GB total · device_map balanced · peak — GB / GPU · wall time **— h**
Hardware (infer)	2x Tesla T4 · peak 10.49 GB / GPU · wall time **3.82 h**
Versions	torch 2.10.0+cu128 · transformers 4.57.6 · peft 0.19.1 · bitsandbytes 0.50.2 · python 3.12.13

2. Training

Metric	Value
Steps completed	20 (max_steps=20, epoch 5.00)
Final train loss	0.9407
Best eval loss	0.9539 at step 20
Eval loss (first → last)	0.9742 → 0.9539
Wall-clock guard	7.5 h — not hit

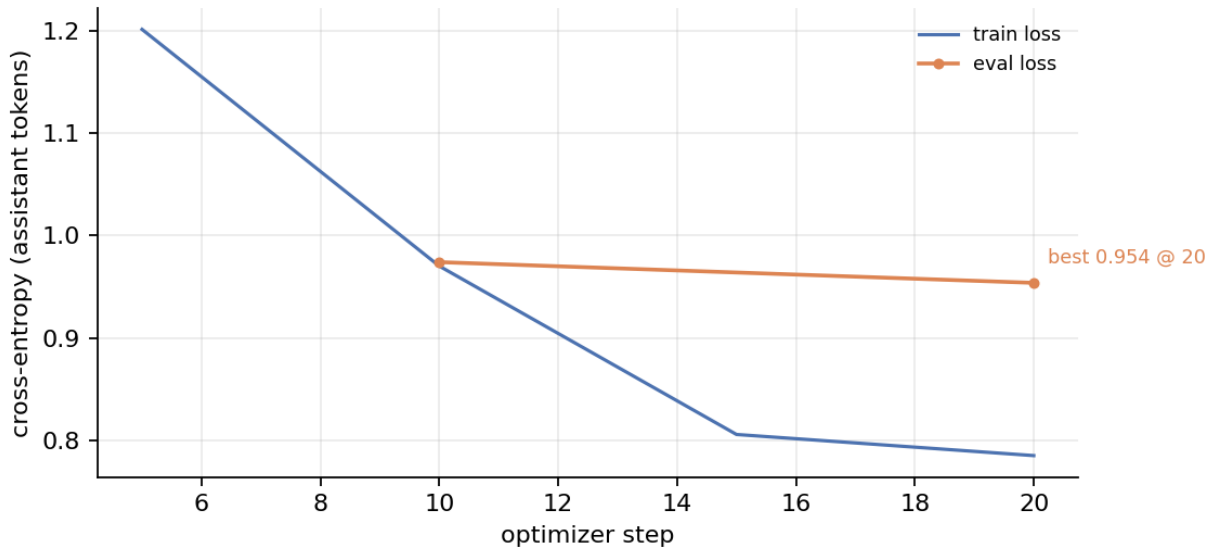


Figure 1 — assistant-token cross-entropy on train (every logging step) and on the 220-item validation set (every eval step). load_best_model_at_end selects the checkpoint at the lowest eval loss for generation.

3. Validation checks (REVIEW2_BASELINE.md §7)

Check	Result	Expectation
Loss report	final train 0.941 · best eval 0.954 @ 20	eval loss decreasing, best step recorded
Length ratio (pred / ref words)	2.114	0.8 – 1.2
Quick ROUGE-L on val subset (n=20)	26.7	sanity gate only
Truncated / empty on val	0 / 0	0 / 0
Adapter reload test	fresh kernel · adapter = checkpoint not mounted · 0 empty outputs	generation from a fresh PeftModel.from_pretrained, not the training process
Zero-shot base control (§7 step 6, same 20 items)	ROUGE-L 21.2 vs adapter 26.7 · len ratio 2.781 vs 2.114 · truncated 0 vs 0	adapter should be more note-like: higher ROUGE-L, ratio nearer 1
ROUGE-L gate · aci_bench val	32.6 — BELOW GATE	≥ 35
ROUGE-L gate · mts_dialog val	20.8 — BELOW GATE	≥ 30

Reload note: generation runs in a separate kernel from training (fresh base + PeftModel.from_pretrained), which is the §7 step 5 reload test itself; non-quantised params upcast to fp32 as in training

4. Test-set generation (REVIEW2_BASELINE.md §8)

Greedy decoding (num_beams=1, do_sample=False, repetition_penalty=1.2, max_new_tokens=1024, 6-gram repeat block on generated tokens). ROUGE F1 × 100 with Porter stemming, computed here as a first look — Person 3's evaluation (BERTScore, coverage, hallucination, negation, numeric consistency) supersedes these. DialogueSOAP is synthetic and is not a headline number.

Split	n	R-1	R-2	R-L	len ratio	gen tok	trunc	empty	status
aci_bench_test_taskB_1	40	53.5	23.1	32.4	1.05	600	1	0	headline
aci_bench_test_taskC_2	40	54.4	23.4	32.3	1.01	614	0	0	headline
aci_bench_test_taskC_3	40	53.2	22.4	31.2	0.97	631	1	0	headline
mts_dialog_test_chat_2023	200	30.4	9.1	24.5	3.31	66	0	0	headline
mts_dialog_test_sum_2023	200	31.6	9.9	24.4	3.13	67	0	0	headline
dialogue_soap_test	250	37.9	15.4	25.6	0.57	212	0	0	synthetic
headline (weighted)	520	36.2	12.6	26.2	2.71	193	2	0	

5. Handoff checklist (REVIEW2_BASELINE.md §10)

	Item	Detail
☐	adapter used for generation == final/adapter_model.safetensors (sha256)	final/ not present locally — run `fetch_kaggle_run.py --stage train --no-wait`
☑	predictions/baseline_qlora/: 6 files, 770 lines, 0 JSON errors	6 files, 770 lines, 0 JSON errors
☑	truncated == true count per split (reported for Person 3 to flag/exclude)	1 / 0 / 1 / 0 / 0 / 0
☑	training_config.json matches trainer_state.json (steps, best_step)	config steps=20 best=20 | state global_step=20
☑	train_manifest sha256 matches kaggle/dataset/baseline_{train,val}.jsonl	verified
☑	final/ adapter reloaded in a fresh kernel and generated on-task output (§7 step 5)	sha256 f431c7434ba6e411 = checkpoint not mounted | matches train_summary: None | empty outputs 0
☑	no test id appears in baseline_train/val.jsonl	0 overlapping ids
☑	environment.txt present	pip freeze captured
☒	final/ adapter present	missing

☑ verified · ☒ failed · ☐ could not be verified from this machine

6. Sample predictions

One item per split, chosen as the record whose ROUGE-L is closest to that split's mean (representative, not cherry-picked). Text is clipped to 900 characters.

aci_bench_test_taskB_1 — aci_bench_test_taskB_1_D2N103-virtscribe **· prompt 1491 tok · generated 518 tok**

Reference	Prediction
CHIEF COMPLAINT Status post mitral valve repair. HISTORY OF PRESENT ILLNESS Mrs. Melissa Sanchez is a 58-year-old female being seen today for a status post mitral valve repair, completed on 08/03/2020. On 09/17/20 we saw Mrs. Sanchez in office, and she was doing well and thus we recommended maintaining her current medications, she is using a reminder app to stay compliant with medications. The patient reports feeling better and overall, pretty good. She denies experiencing new symptoms. She is still having chest pain intermittently, and her breathing “gets shallow” which results in her “slowing down” and decreasing her exertion. An episode of shortness of breath and chest pains may last a few minutes and will not recur for a few weeks. PAST HISTORY Medical Mitral regurgitation. Atrial fibrillation. Diabetes Type II. Asthma. Surgical Mitral valve repair 08/03/2020. CURRENT MEDICA …	CHIEF COMPLAINT Status post mitral valve repair. HISTORY OF PRESENT ILLNESS Melissa Sanchez is a pleasant 61-year-old female who presents to clinic today for a follow-up appointment after undergoing mitral valve repair on August 3rd, 2020. The patient reports that she feels "pretty good" currently; however, she continues to experience occasional episodes of chest pain along with shortness of breath. She states these episodes resolve within several minutes once she slows down. These events occur approximately two to three times per month. Her medications include Coumadin 4 mg daily, Lasix 40 mg daily, and Atenolol 50 mg daily. She denies experiencing any adverse reactions related to her prescribed medications except for initial irritability associated with the use of Atenolol, which seems less bothersome over time. REVIEW OF SYSTEMS Cardiovascular: Reports dyspnea. Respiratory: Denie …

aci_bench_test_taskC_2 — aci_bench_test_taskC_2_D2N151-aci **· prompt 3115 tok · generated 648 tok**

Reference	Prediction
CHIEF COMPLAINT Right elbow pain. HISTORY OF PRESENT ILLNESS Teresa Sullivan is a pleasant 33-year-old female who presents to the clinic today for the evaluation of right elbow pain. Her pain began 4 days ago, without any improvement. She denies any specific injury; however, she has recently increased her usage of a pottery wheel in creating a bowl for her son. She is currently unable to work with the pottery wheel secondary to pain. Her pain is in the medial aspect of her elbow. She notes it radiates medially down her arm as well. The patient rates her pain level as 7 out of 10 at its worst. She reports the pain wakes her at night if she rolls over. She denies ibuprofen or ice provide pain relief. The patient denies any previous elbow injuries. The patient has a history of hypertension. She states she has been checking her blood pressures regularly. She adds that her blood pressures …	CHIEF COMPLAINT Right elbow pain. HISTORY OF PRESENT ILLNESS Teresa Smith is a pleasant 65-year-old female who presents to the clinic today for the evaluation of her right elbow pain. The onset of her symptoms began approximately 4 days ago. She states she has been engaging in activities involving pottery which involves heavy lifting and repetitive motion. Her pain is localized to the medial aspect of her right elbow. It radiates distally into her right forearm. She denies hitting her elbow on anything. She rates her pain level as a 7/10. She adds that she experiences pain upon rolling onto her right side at night. She reports that she has taken ibuprofen without relief. She notes that she applied ice initially but stopped after a few applications. She continues to perform light walking exercises. The patient endorses hypertension and diabetes mellitus. She takes Lisinopril 10 mg onc …

aci_bench_test_taskC_3 — aci_bench_test_taskC_3_D2N189-aci · prompt 1658 tok · generated 606 tok

Reference	Prediction
CHIEF COMPLAINT Follow up. MEDICAL HISTORY Patient reports an myocardial infarction. SURGICAL HISTORY Patient reports stent placement in the right coronary artery. SOCIAL HISTORY Patient is a fan of the Denver Nuggets basketball team. He also enjoys swimming. MEDICATIONS Patient reports taking Brilinta twice a day every 12 hours, aspirin 81 mg at night, Crestor 40 mg once a day, and Toprol 50 mg once a day. REVIEW OF SYSTEMS Cardiovascular: Denies chest pain. Respiratory: Reports shortness of breath with exertion. Skin: Denies redness or infection in the right wrist. VITALS Blood Pressure: 127/60. PHYSICAL EXAM Neck - General Examination: No jugular venous distention. No carotid bruits. Respiratory - Auscultation of Lungs: Clear bilaterally. Cardiovascular - Auscultation of Heart: Slight 3/6 systolic ejection murmur heard at the left base. No gallops or rubs. Musculoske …	CHIEF COMPLAINT Right coronary artery stenting due to acute inferior myocardial infarction. HISTORY OF PRESENT ILLNESS Betty Johnson is a pleasant 65-year-old female who presents today for a post-hospital visit following her recent admission for acute inferior myocardial infarciton treated with percutaneous transluminal angioplasty (PTCA) and placement of a bare metal stent in the right coronary artery. She reports she has been doing "okay" since discharge approximately 2 weeks ago. The patient states she continues to experience dyspnea on exertion; however, she notes she feels better than before her current illness. Her right wrist appears to be healing nicely and she denies any signs of infection. She confirms she takes Brilinta twice daily and baby aspirin nightly. She adds that she eats healthy during the week and consumes pizza occasionally on the weekend. She expresses concern r …

mts_dialog_test_chat_2023 — mts_dialog_test_chat_2023_195 · prompt 335 tok · generated 70 tok

Reference	Prediction
Briefly, the patient is a very pleasant 72-year-old female with previous history of hypertension and also recent diagnosis of C. diff for which she was admitted here in 5/2009, who presents to the hospital on 6/18/2009 with abdominal pain, cramping, and persistent diarrhea. After admission, she had a CT of the abdomen done, which showed evidence of diffuse colitis and she was started on IV Flagyl and also on IV Levaquin. She was also placed on IV Reglan because of nausea and vomiting. In spite of the above, her white count still continues to be elevated today. On questioning the patient, she states the nausea and vomiting has resolved, but the diarrhea still present, but otherwise denies any other specific complaints except for some weakness.	HISTORY OF PRESENT ILLNESS The patient is a pleasant 72-year-old female who presents today for follow-up care after her hospitalization last month. She reports she has been doing better with resolution of her nausea and vomiting; however, she continues to experience persistent diarrhea which causes weakness. The patient denies current gastrointestinal distress at present time.

mts_dialog_test_sum_2023 — mts_dialog_test_sum_2023_199 · prompt 158 tok · generated 28 tok

Reference	Prediction
Remarkable only for hypertension in her father. Her mother died in a motor vehicle accident.	FAMILY HISTORY Mother deceased due to motor vehicle accident; otherwise asymptomatic. Father hypertension. SOCIAL HISTORY Employed full-time.

dialogue_soap_test — dialogue_soap_test_242 · prompt 766 tok · generated 214 tok

Reference	Prediction
S: The patient was referred for evaluation of a mixed ground-glass opacity lesion in the right upper lung. The patient reports feeling fine and has not noticed any symptoms. There is no history of similar issues or relevant medical conditions mentioned. O: Physical examination showed no remarkable findings. Chest CT revealed a 34 mm x 20 mm partial solid nodule with spicule formation and pleural indentation at the right S3, with a consolidation tumor ratio of 30%. PET-CT indicated an SUV of 1.9 with no suspicious metastasis. Additional findings included a Kommerell diverticulum at the origin of the left subclavian artery and a right-sided aortic arch from 3D CT reconstruction. Echocardiography showed no cardiac abnormalities. Surgical intervention involved a right upper lobectomy using a three-port VATS approach. Histology confirmed invasive carcinoma. Radical mediastinal lymph node diss …	S: The patient reports she feels well. O: Physical Examination: No respiratory distress; lungs clear bilaterally without wheezing, rales, rhonchi, or decreased breath sounds. Chest Computed Tomography Scan Reveals Partial Solid Nodules With Spicule Formation And Pleural Indentation At Right S3. Consolidation Tumor Ratio Was 30%. PET/CT Shows Standard Uptake Value Of Lesion Is 1.9 Without Suspicious Metastases. Three-Dimensional Reconstruction Demonstrates Left Subclavian Artery Diverticulum Originating From Kommerell’s Diverticulum And Right-Sided Arch. Echocardiogram Normal. A: Invasive Carcinoma. P: We discussed her current condition in detail and reviewed all test results. She underwent a right upper lobectomy via video-assisted thoracic surgery (VATS) due to suspicion of malignancy. Radical mediastinal lymphadenectomy was also performed. Her post-op course has been uneventful …

7. Environment

accelerate 1.13.0 · bitsandbytes 0.50.2 · liger_kernel 0.8.2 · peft 0.19.1 · rouge_score 0.1.2 · torch 2.10.0+cu128 · transformers 4.57.6

Run directory: outputs\kaggle_run2_smoke_adapter\outputs\baseline_qlora_llama31_8b. Predictions and adapter stay under outputs/ (git-ignored; corpora licences forbid redistribution). This PDF, metrics.json, training_config.json and trainer_state.json are committed under reports/.